Hands-On-Activity No. 3.1

Hands-on Activity 3.1: Control Structures (part 2)

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming logic and Design	**Date Performed: July 23, 2026**
Section: CPE11S2	**Date Submitted: July 24, 2026**
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1. Objective(s)

This activity aims to develop programming skills using selection and repetition structures.

2. Intended Learning Outcomes

2.1 Understand and use conditions and conditional executions with the if/*else* selection structure

2.2 Understand and use conditions and conditional executions with the *while* selection structure

2.3 Understand and use counter controlled repetition and sentinel controlled repetition.

3. Materials and Equipment

Internet connection
Personal Computer

4. Procedure

5. Output

1. Develop a C++ program that will determine if a department store customer has exceeded the credit limit on a charge account. For each customer, the following facts are available:

1. Account number
2. Balance at the beginning of the month
3. Total of all items charged by this customer this month
4. Total of all credits applied to this customer's account this month
5. Allowed credit limit

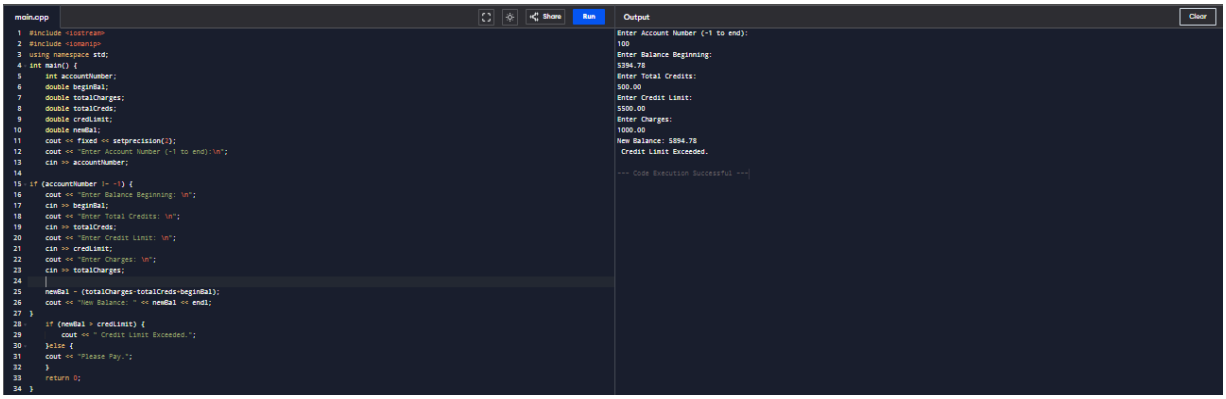
The program should input each of these facts, calculate the new balance (=*beginning balance +charges -credits),* and determine if the new balance exceeds the customer's credit limit. For those customers whose credit limit is exceeded, the program should display the customer's account number, credit limit, new balance and the message "Credit limit exceeded."

Answer:

```
#include <iostream>
#include <iomanip>
using namespace std;
int main() {
   int accountNumber;
   double beginBal;
   double totalCharges;
   double totalCreds;
   double credLimit;
   double newBal;
   cout << fixed << setprecision(2);
   cout << "Enter Account Number (-1 to end):\n";
   cin >> accountNumber;
```

```
        if (accountNumber != -1) {
            cout << "Enter Balance Beginning: \n";
            cin >> beginBal;
            cout << "Enter Total Credits: \n";
            cin >> totalCreds;
            cout << "Enter Credit Limit: \n";
            cin >> credLimit;
            cout << "Enter Charges: \n";
            cin >> totalCharges;

            newBal = (totalCharges-totalCreds+beginBal);
            cout << "New Balance: " << newBal << endl;
        }
            if (newBal > credLimit) {
                cout << " Credit Limit Exceeded.";
            }else {
            cout << "Please Pay.";
            }
            return 0;
        }
```



2. Because of the price of gasoline, drivers are concerned with the mileage obtained by their automobiles. One driver has kept track of several tankfuls of gasoline by recording miles driven and gallons used for each tankful. Develop a program that will input the miles driven and gallons used for each tankful. The program should calculate and display the miles per gallon obtained for each tankful. After processing all input information, the program should calculate and print the combined miles per gallon obtained for all tankfuls.

Answer:

```
#include <iostream>
#include <iomanip>
using namespace std;
        int main() {
            double literUsed;
            double KpL;
            double kilometers;
            double totalKilometers = 0.00;
```

```
    double totalLiters = 0.00;

    cout << fixed << setprecision(2);

    cout << "Enter the kilometer driven (-1 to end):\n";
    cin >> kilometers;

  while (kilometers != -1) {
    cout << "Enter liters used: \n";
    cin >> literUsed;

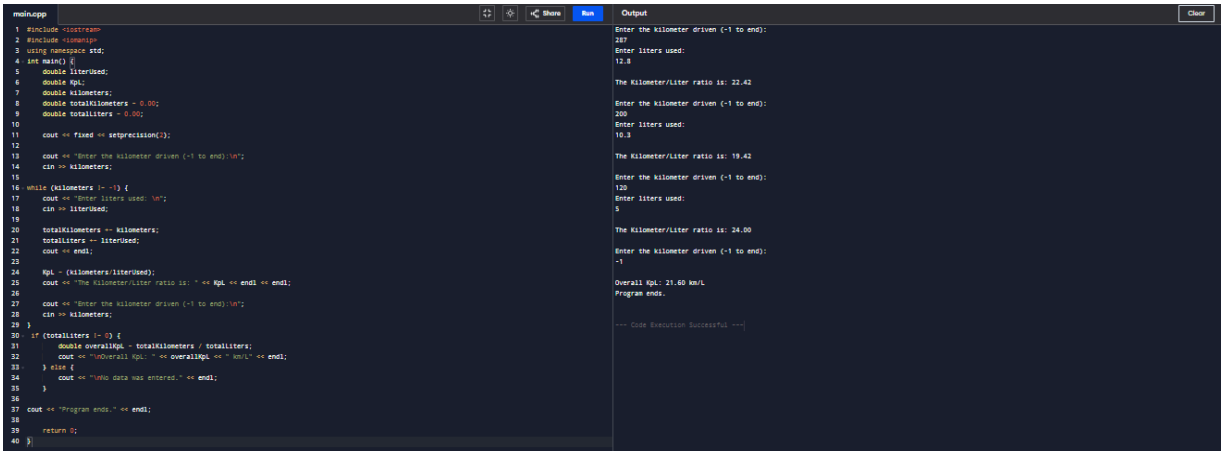
    totalKilometers += kilometers;
    totalLiters += literUsed;
    cout << endl;

    KpL = (kilometers/literUsed);
    cout << "The Kilometer/Liter ratio is: " << KpL << endl << endl;

    cout << "Enter the kilometer driven (-1 to end):\n";
    cin >> kilometers;
  }
   if (totalLiters != 0) {
       double overallKpL = totalKilometers / totalLiters;
       cout << "\nOverall KpL: " << overallKpL << " km/L" << endl;
    } else {
       cout << "\nNo data was entered." << endl;
    }

  cout << "Program ends." << endl;

    return 0;
  }
```



3. Create a program that will calculate the cost of sending a small parcel. The post office charges P5.00 for the first 300g, and P2.00 for every 100g thereafter (rounded up), up to a maximum weight of 1000g.

Enter the name of the item/s:
Enter the weight of the item (grams):

Answer:

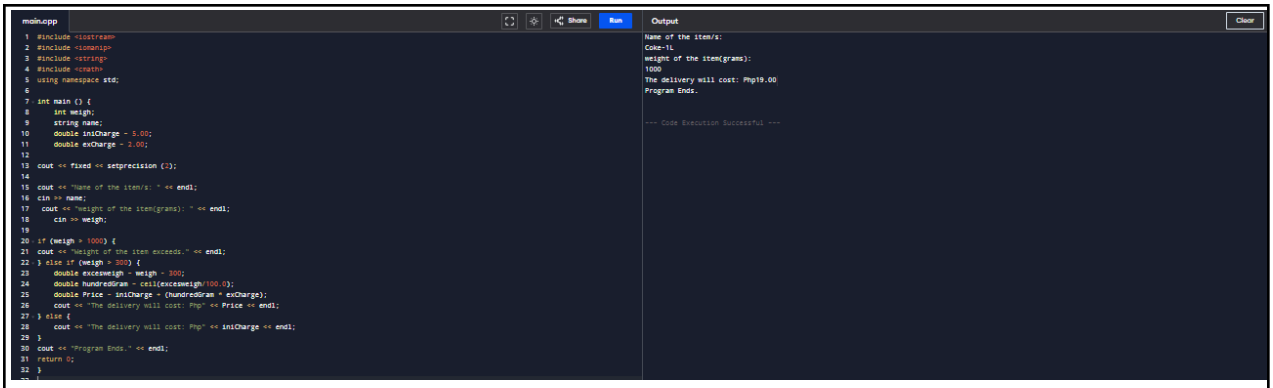
```
#include <iostream>
#include <iomanip>
#include <string>
#include <cmath>
using namespace std;

        int main () {
           int weigh;
           string name;
           double iniCharge = 5.00;
           double exCharge = 2.00;

        cout << fixed << setprecision (2);

        cout << "Name of the item/s: " << endl;
        cin >> name;
         cout << "weight of the item(grams): " << endl;
           cin >> weigh;

        if (weigh > 1000) {
        cout << "Weight of the item exceeds." << endl;
        } else if (weigh > 300) {
           double excesweigh = weigh - 300;
           double hundredGram = ceil(excesweigh/100.0);
           double Price = iniCharge + (hundredGram * exCharge);
           cout << "The delivery will cost: Php" << Price << endl;
        } else {
           cout << "The delivery will cost: Php" << iniCharge << endl;
        }
        cout << "Program Ends." << endl;
        return 0;
        }
```



4. Write a program that displays a menu for simple conversion such as the following:

- **(1) cm - inches**
- **(2) inches - cm**
- **(3) feet - meter**
- **(4) meter - feet**

Answer:

```
#include <iostream>
#include <iomanip>
using namespace std;

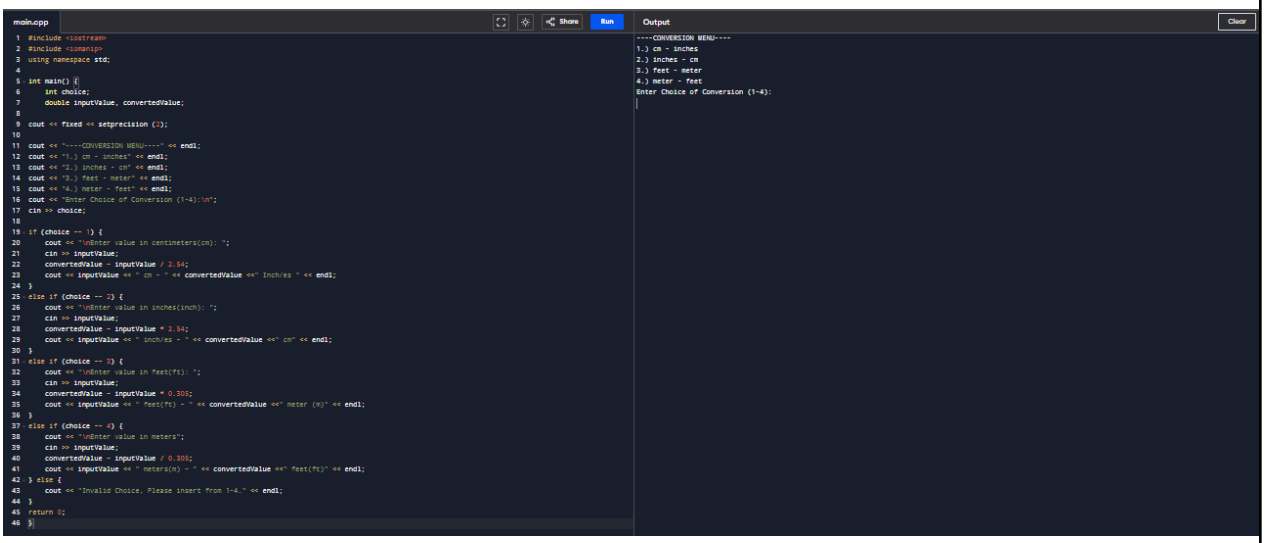
int main() {
   int choice;
   double inputValue, convertedValue;

cout << fixed << setprecision (2);

cout << "----CONVERSION MENU----" << endl;
cout << "1.) cm - inches" << endl;
cout << "2.) inches - cm" << endl;
cout << "3.) feet - meter" << endl;
cout << "4.) meter - feet" << endl;
cout << "Enter Choice of Conversion (1-4):\n";
cin >> choice;

if (choice == 1) {
   cout << "\nEnter value in centimeters(cm): ";
   cin >> inputValue;
   convertedValue = inputValue / 2.54;
   cout << inputValue << " cm = " << convertedValue <<" Inch/es " << endl;
}
else if (choice == 2) {
   cout << "\nEnter value in inches(inch): ";
   cin >> inputValue;
   convertedValue = inputValue * 2.54;
   cout << inputValue << " inch/es = " << convertedValue <<" cm" << endl;
}
else if (choice == 3) {
```

```
        cout << "\nEnter value in feet(ft): ";
        cin >> inputValue;
        convertedValue = inputValue * 0.305;
        cout << inputValue << " feet(ft) = " << convertedValue <<" meter (m)" << endl;
    }
    else if (choice == 4) {
        cout << "\nEnter value in meters";
        cin >> inputValue;
        convertedValue = inputValue / 0.305;
        cout << inputValue << " meters(m) = " << convertedValue <<" feet(ft)" << endl;
    } else {
        cout << "Invalid Choice, Please insert from 1-4." << endl;
    }
    return 0;
    }
```



5. Write a program that displays a menu for simple computation of formula such as the following:

- (1) Area of circle, will ask for radius from user
- (2) Area of rectangle, will ask for L and W
- (3) Area of triangle, will ask for B and H
- (4) Area of square - feet, will ask for S

answer:

```
#include <iostream>
#include <iomanip>
#include <cmath>
    using namespace std;

    int main() {
        int choice;
        char contrast;
        float radius, length, width, base, height, side, area;

        cout << fixed << setprecision (2);
```

```
do {
cout << "----Geometry Computation Menu----" << endl;
cout << "1.) Area of Circle" << endl;
cout << "2.) Area of Rectangle" << endl;
cout << "3.) Area of Triangle" << endl;
cout << "4.) Area of Square" << endl;
cout << "Enter chosen number (1-4):\n";
cin >> choice;

switch (choice) {
   case 1:
      cout << "Enter radius of the circle: ";
      cin >> radius;
      area = M_PI * radius * radius;
      cout << "Area of the Circle: " << area << endl;
            break;

   case 2:
      cout << "Enter Length (L): ";
      cin >> length;
      cout << "Enter Width (W): ";
      cin >> width;
      area = length * width;
      cout << "Area of the Rectangle: " << area << endl;
      break;

   case 3:
      cout << "Enter Base (B): ";
      cin >> base;
      cout << "Enter Height (H): ";
      cin >> height;
      area = 0.5f * base * height;
      cout << "Area of the Triangle: " << area << endl;
      break;

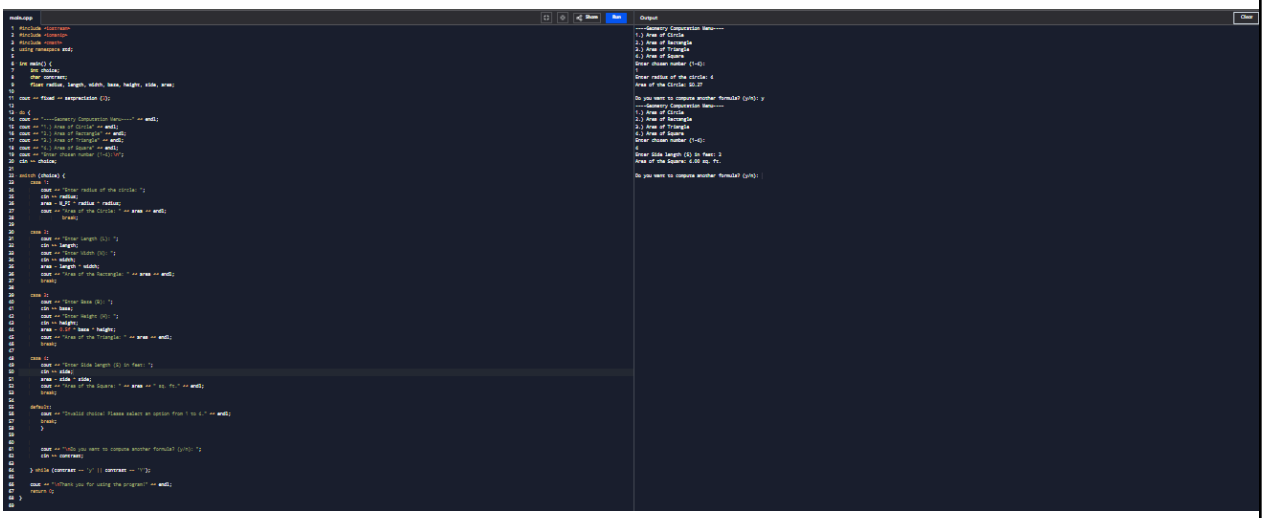
   case 4:
      cout << "Enter Side length (S) in feet: ";
      cin >> side;
      area = side * side;
      cout << "Area of the Square: " << area << " sq. ft." << endl;
      break;

   default:
      cout << "Invalid choice! Please select an option from 1 to 4." << endl;
      break;
      }


      cout << "\nDo you want to compute another formula? (y/n): ";
      cin >> contrast;

   } while (contrast == 'y' || contrast == 'Y');

   cout << "\nThank you for using the program!" << endl;
   return 0;
}
```



```
#include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;

int main() {
    int choice;
    char continue;
    float radius, length, width, base, height, side, area;

cout << fixed << setprecision (2);

do {
cout << "----Geometry Computation Menu----" << endl;
cout << "1.) Area of Circle" << endl;
cout << "2.) Area of Rectangle" << endl;
cout << "3.) Area of Triangle" << endl;
cout << "4.) Area of Square" << endl;
cout << "Enter chosen number (1-4): ";
cin >> choice;

switch (choice) {
    case 1:
        cout << "Enter radius of the circle: ";
        cin >> radius;
        area = 0.5*3.14 * radius * radius;
        cout << "Area of the Circle: " << area << endl;
            break;

    case 2:
        cout << "Enter Length (L): ";
        cin >> length;
        cout << "Enter Width (W): ";
        cin >> width;
        area = length * width;
        cout << "Area of the Rectangle: " << area << endl;
        break;

    case 3:
        cout << "Enter Base (B): ";
        cin >> base;
        cout << "Enter Height (H): ";
        cin >> height;
        area = 0.5f * base * height;
        cout << "Area of the Triangle: " << area << endl;
        break;

    case 4:
        cout << "Enter Side length (S) in feet: ";
        cin >> side;
        area = side * side;
        cout << "Area of the Square: " << area << " sq. ft." << endl;
        break;

    default:
        cout << "Invalid choice! Please select an option from 1 to 4." << endl;
        break;
        }

        cout << "\nDo you want to compute another formula? (y/n): ";
        cin >> continue;

    } while (continue == 'y' || continue == 'Y');

    cout << "\nThank you for using the program!" << endl;
    return 0;
}
```

```
----Geometry Computation Menu----
1.) Area of Circle
2.) Area of Rectangle
3.) Area of Triangle
4.) Area of Square
Enter chosen number (1-4):
1
Enter radius of the circle: 4
Area of the Circle: 50.27

Do you want to compute another formula? (y/n): y
----Geometry Computation Menu----
1.) Area of Circle
2.) Area of Rectangle
3.) Area of Triangle
4.) Area of Square
Enter chosen number (1-4):
4
Enter Side length (S) in feet: 3
Area of the Square: 6.00 sq. ft.

Do you want to compute another formula? (y/n):
```

6. Conclusion

Learned about creating some basic interactable codes the can calculate.

7. Rubric